



PES UNIVERSITY, Bengaluru
Department of Computer Science and Engineering
B. Tech (CSE) – 5th Semester – Aug-Dec 2023

UE21CS341A - Software Engineering
SOFTWARE REQUIREMENTS SPECIFICATION
HAPPY JOURNEY!

Your All-in-One Travel Companion App 🌐✈️📱 – Explore, Plan, and Connect with
Fellow Travelers!"

VERSION 1

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Date Created : 09-10-2023

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Revision History

Name	Date	Reason for Change	Version

1. Introduction

1.1 Purpose

This document outlines the software requirements for the "Happy Journey" travel companion application, hereafter referred to as "the product." This SRS details the functional and non-functional requirements for the initial release of the product, version 1.0.

The scope of this SRS encompasses the web-based "Happy Journey" application. It covers the core features and functionalities required to deliver a comprehensive travel companion experience to users. This document describes the primary subsystems and their interactions within the web-based product.

1.2 Intended Audience

This Software Requirements Specification (SRS) is intended for various stakeholders involved in the development, implementation, and quality assurance of the "Happy Journey" web-based application. The document is organized to cater to the following types of readers:

- **Developers:** Developers will use this SRS to understand the detailed functional and technical requirements of the application. It provides insights into the architecture, data flows, and system behavior required for development.
- **Project Managers:** Project managers will find this document valuable for project planning and management. It outlines project scope, objectives, timelines, and resource requirements, facilitating effective project oversight.
- **Testers:** Testers will refer to this SRS to create test plans, test cases, and perform validation and verification of the application against specified requirements. It provides the basis for quality assurance and testing efforts.
- **UI/UX Designers:** UI/UX designers will use this document to understand the design requirements, user interface elements, and user experience expectations to create a visually appealing and user-friendly application.

- Marketing and Sales Teams: Marketing and sales staff can gain insights into the key features and benefits of the application, helping them promote and communicate the value proposition to potential users and clients.
- Users: While not the primary audience, users may reference this document to understand the capabilities and functionality of the "Happy Journey" application. However, user documentation will be provided separately for their convenience.

Organization of this SRS:

The remainder of this SRS is organized as follows:

Section 2: "Overall Description" provides an overview of the product's perspective, functions, user classes, operating environment, and design constraints.

Section 3: "Specific Requirements" details the specific functional and non-functional requirements of the application, including use cases, system features, and interfaces.

Section 4: "External Interface Requirements" outlines user interfaces, software interfaces, and communication interfaces.

Section 5: "System Features" describes individual features of the application in detail.

Section 6: "Non-Functional Requirements" presents performance, security, and other non-functional requirements.

Section 7: "Appendices" includes any supplementary information, references, and supporting documents.

This organization ensures that each stakeholder can easily locate the information relevant to *their role and responsibilities*.

1.3 Product Scope

Product Description:

The "Happy Journey" web-based application is a comprehensive travel companion designed to simplify and enhance the travel experience for users. It serves as a one-stop platform for travel planning, exploration, and community engagement.

Purpose:

The purpose of the "Happy Journey" application is to provide users with the following benefits:

- **Streamlined Travel Planning:** Users can efficiently plan their trips by accessing information on restaurants, attractions, and real-time weather conditions at their destinations.
- **Enhanced Exploration:** The application offers detailed insights into local attractions and points of interest, enriching users' travel experiences.
- **Travel Booking:** Users can conveniently book flights, hotels, and travel packages directly through the app, ensuring a seamless booking process.
- **Community Interaction:** "Happy Journey" fosters a sense of community by allowing users to connect with fellow travelers, share experiences, and discover travel partners.

Objectives:

The objectives of the "Happy Journey" application are as follows:

- To provide users with a user-friendly, informative, and engaging platform for travel planning and exploration.
- To offer a seamless travel booking experience, promoting convenience and efficiency.
- To facilitate connections and interactions among travelers, creating a vibrant and supportive travel community.

Alignment with Corporate Goals:

The development of the "Happy Journey" application aligns with our corporate goals of:

Expanding our digital presence in the travel industry.

Enhancing the user experience and satisfaction of travelers.

Driving revenue through bookings and partnerships with travel service providers.

Fostering a strong and engaged user community.

For a more detailed vision and scope of the project, please refer to the separate "Happy Journey Vision and Scope" document.

1.4 References

User Interface Style Guide:

Title: "Happy Journey Synopsis"

Version: 1.0

Date: Sep 20,2023

Source/Location: [SE23 P01 T13 Happy Journey!.pdf](#)

Title: "Happy Journey Project Plan"

Version: 1.0

Date: October 1,2023

Source/Location: [SE23-5J P02 T13 Happy Journey!.pdf](#)

System Requirements Specification:

Title: "Happy Journey System Requirements Specification"

Version: 1.0

Date: August 20, 2023

Source/Location: [SE23-5J P03 T13 Happy Journey!.pdf](#)

Vision and Scope Document:

Title: "Happy Journey Vision and Scope"

Author: Your Company Name

Version: 1.2

Date: July 5, 2023

Source/Location: [SE23-5J P03 T13 Happy Journey!.pdf](#)

Use Case Documents:

Title: "Happy Journey Use Cases"

Author: Your Company Name

Version: 1.0

Date: September 1, 2023

Source/Location: [1.jpg](#)

<https://developers.google.com/docs/api/reference/rest>

<https://cloud.google.com/>

<https://rapidapi.com/>

<https://www.mongodb.com/>

<https://react.dev/>

<https://nodejs.org/en>

<https://expressjs.com/>

2. Overall Description

2.1 Product Perspective

The "Happy Journey" web-based application is a new, self-contained product developed to provide users with a comprehensive and user-friendly travel companion experience. This product is not part of an existing product family but has been conceived as a standalone solution to meet the evolving needs of modern travelers.

1. Origin:

The idea for the "Happy Journey" application originated from a market analysis that identified a growing demand for a centralized platform that simplifies travel planning, enhances exploration experiences, and fosters a vibrant travel community. Recognizing this opportunity, our organization embarked on the development of this innovative travel companion.

2. Relationship to Larger System:

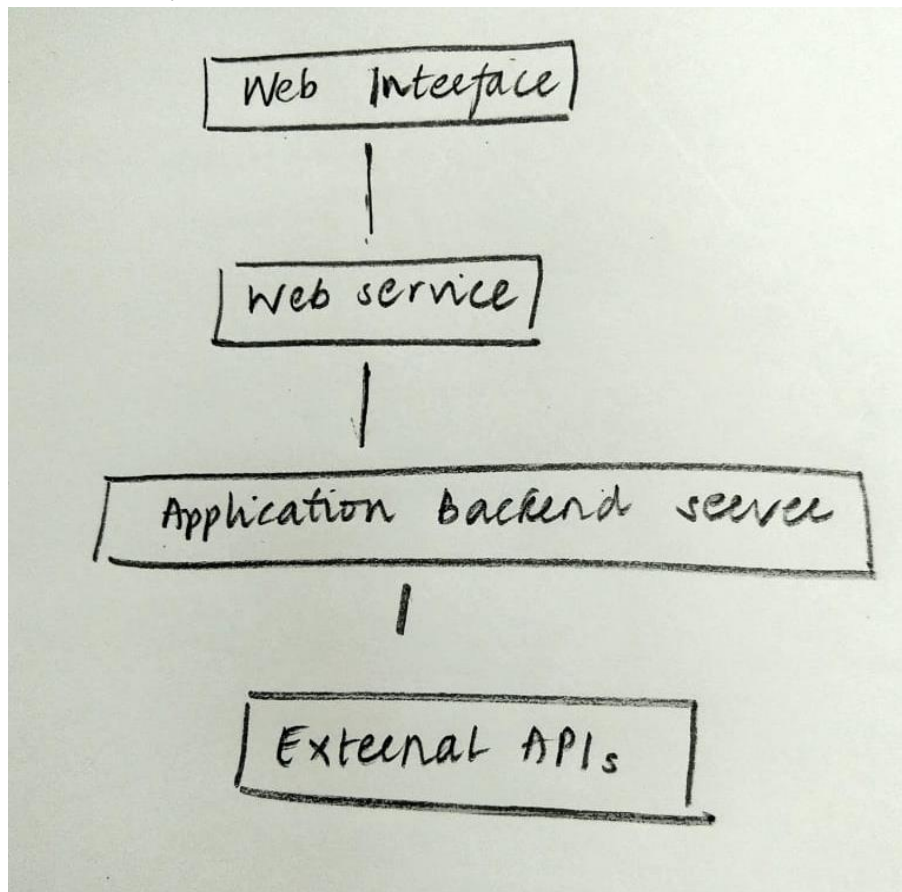
The "Happy Journey" application is designed as a self-contained system that does not depend on or integrate with other existing systems or products within our organization. It is a standalone solution that provides a complete travel planning and exploration experience within a single platform.

3. System Components:

While "Happy Journey" is self-contained, it consists of several major components:

- **User Interface (UI):** This component provides the front-end interface that users interact with. It includes web pages for travel planning, user profiles, restaurant discovery, attractions, travel bookings, and community interaction.
- **Backend Services:** The backend services handle user data, authentication, booking processing, and integration with external data sources, such as real-time weather updates.

- Database: The database stores user profiles, travel data, restaurant information, attraction details, and other relevant data to ensure smooth functionality.
- External Interfaces: "Happy Journey" interacts with external services for travel booking, real-time weather updates, and social media integration to support community features.

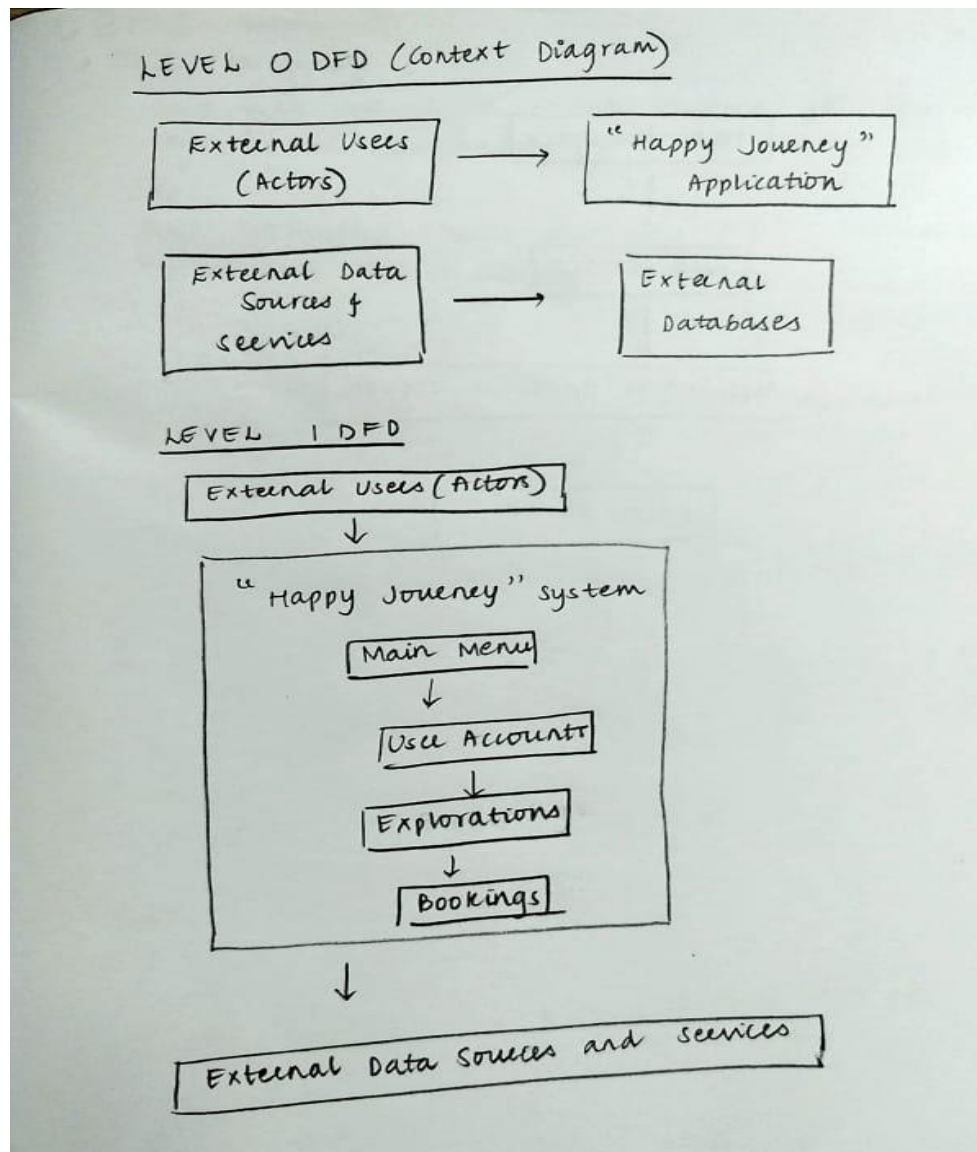


2.2 Product Functions

The "Happy Journey" application provides users with a wide range of functions to simplify travel planning, enhance exploration, and foster a vibrant travel community. The major product functions include:

- User Registration and Login: Users can create accounts and log in securely to access personalized features.
Has a very beautiful, animated and responsive login page.

- Restaurant Discovery: Users can discover nearby restaurants, view menus, read reviews, and make reservations.
- Attractions and Points of Interest: Users can explore attractions, view details, check opening hours, and plan visits.
- Travel Booking: Users can book flights, hotels, and travel packages directly through the app.
- Search for Travel Partners: Users can search for potential travel partners based on shared interests, travel plans, and preferences.
- Community Interaction: Users can connect with fellow travelers, share experiences, and engage in discussions.
- Real-Time Weather Updates: Users receive real-time weather forecasts for their travel destinations.
- User Notifications: Users receive notifications for booking confirmations, community activity, and important updates.
- User Dashboard: Users have access to a personalized dashboard that displays relevant information and recommendations.



2.3 User Classes and Characteristics

The "Happy Journey" application is designed to cater to various user classes, each with distinct characteristics and needs. Understanding these user classes is essential for tailoring the user experience and meeting specific requirements. The anticipated user classes include:

- Travel Enthusiasts:

- Characteristics: These users have a high frequency of app use and engage with all product functions. They are experienced travelers, tech-savvy, and value detailed travel planning.
- Requirements: Access to all features, personalized recommendations, and advanced customization options.
- Casual Travelers:

Characteristics: Casual travelers use the app periodically for basic travel planning. They may not be as experienced and prefer simplicity.

Requirements: Easy-to-use interface, quick access to basic travel information, and simplified booking options.
 - Foodies and Restaurant Lovers:

Characteristics: These users primarily use the app for restaurant discovery and dining experiences. They are interested in culinary exploration.

Requirements: Robust restaurant search and review features, reservation functionality, and cuisine-specific recommendations.
 - Explorers and Sightseers:

Characteristics: Explorers focus on discovering attractions and points of interest. They seek detailed information about places to visit.

Requirements: Comprehensive attraction listings, detailed descriptions, and location-based search.
 - Solo Travelers:

Characteristics: Solo travelers often use the app to find travel partners or join travel groups. They value social interaction during their trips.

Requirements: User-friendly travel partner search functionality, secure communication options, and group trip planning features.

- Community Contributors:

Characteristics: These users actively engage in community discussions, share travel experiences, and contribute to user-generated content.

Requirements: User-friendly content creation and sharing tools, engagement metrics, and moderation capabilities for administrators.

- Administrators:

Characteristics: Admins are responsible for managing user accounts, ensuring content quality, and monitoring community activity.

Requirements: Admin panel for user management, content moderation tools, and activity monitoring features.

2.4 Operating Environment

The "Happy Journey" application operates within a specific environment to ensure optimal performance and compatibility. The key components of this operating environment include:

- Hardware Platform:

The application is designed to run on standard web browsers, making it accessible from various devices, including desktop computers, laptops, tablets, and smartphones.

The hardware requirements are minimal, allowing users with diverse devices to access the application smoothly.

- Operating Systems and Versions:

The web-based application is compatible with a wide range of operating systems, including but not limited to:

- Windows (versions 7 and above)

- macOS (all recent versions)
- Linux (various distributions)
- iOS (for mobile devices)
- Android (for mobile devices)

The application's web interface is designed to be responsive and adaptable to different screen sizes and resolutions.

- Web Browsers:

"Happy Journey" supports major web browsers to ensure broad accessibility, including:

- Google Chrome
- Mozilla Firefox
- Apple Safari
- Microsoft Edge
- Other modern and up-to-date browsers

The application leverages web standards and best practices for cross-browser compatibility.

- Software Components and Dependencies:

The application interacts with external services and APIs for specific functionalities:

Travel booking services for flight and hotel reservations.

Real-time weather data providers for weather updates.

Social media integration for community engagement.

API integration and data exchange are performed securely using industry-standard protocols (e.g., RESTful APIs).

- Additional Considerations:

Users are recommended to have a stable internet connection to access real-time data and functionalities seamlessly.

Browser cookies and local storage may be used to enhance the user experience, but the application remains functional with these features disabled.

"Happy Journey" is designed to be platform-agnostic, ensuring accessibility across a variety of hardware and software environments. Compatibility with standard web browsers and commonly used operating systems guarantees a wide user base.

2.5 Design and Implementation Constraints

The development and implementation of the "Happy Journey" application will be subject to various constraints and considerations, including:

- **Corporate Policies and Regulations:**

The project must adhere to all corporate policies, including data privacy and security regulations, code of conduct, and branding guidelines.

- **Hardware Limitations:**

The application is designed to be lightweight and should operate efficiently on standard hardware configurations to accommodate users with various devices and internet connections.

Memory and processing requirements should be optimized to ensure smooth performance, especially on mobile devices.

- **Interfaces to Other Applications:**

The application will interact with external services for travel booking, real-time weather updates, and social media integration.

API integration and data exchange must follow the protocols and specifications provided by these external services.

- **Technology Stack:**

The development team is constrained to use specific technologies and tools chosen for this project, including:

Front-end technologies: HTML5, CSS3, JavaScript, and a modern web framework (e.g., React).

Back-end technologies: Node.js, Express.js, and a relational database management system (e.g., PostgreSQL).

Version control system: Git and GitHub for code management.

The choice of these technologies is based on compatibility, familiarity, and efficiency.

- Security Considerations:

The application must adhere to strict security standards to protect user data and prevent unauthorized access.

Encryption protocols (e.g., HTTPS) must be implemented for secure data transmission.

User authentication and authorization mechanisms must be robust and regularly updated to address potential security threats.

- Design Conventions and Programming Standards:

The development team is required to follow established design conventions and programming standards.

Code must be well-documented, maintainable, and adhere to best practices.

Quality assurance and testing processes must be in place to identify and rectify any deviations from standards.

- Language Requirements:

The application will support multiple languages for a global user base, with a focus on English as the primary language.

Language localization will be required, including translation and cultural adaptation of content.

- Parallel Operations:

The development team may be working on multiple features or components in parallel, necessitating effective project management and coordination.

- Maintenance and Support:

The customer's organization will be responsible for maintaining the delivered software, including updates, bug fixes, and ongoing support.

These constraints and considerations will guide the development process, ensuring that the "Happy Journey" application is developed in compliance with corporate policies, performance requirements, security standards, and design conventions while providing an optimal user experience.

2.6 Assumptions and Dependencies

➤ Assumptions:

- **Third-Party Services:** The project assumes that third-party services, such as travel booking APIs and real-time weather data providers, will remain available and accessible throughout the project's lifecycle. Changes or interruptions in these services could impact the application's functionality.
- **Data Accuracy:** The accuracy of data obtained from external sources, including restaurant information, attraction details, and weather updates, is assumed but cannot be guaranteed. The application will rely on the quality of data provided by external providers.
- **User Internet Connectivity:** The assumption is that users will have stable internet connectivity to access real-time data, including weather updates and community interactions. The app's performance may be affected in low or unreliable network conditions.
- **Language Localization:** It is assumed that language localization efforts, including translation and cultural adaptation, will adequately address the needs of users in various regions. Localization effectiveness may vary.
- **Dependencies:**

Third-Party APIs: The project is dependent on the continued availability and reliability of third-party APIs for travel booking, real-time weather updates, and social media

integration. Any changes or disruptions in these APIs may require adjustments to the application.

- **Development Environment:** The project relies on specific development tools and environments, including programming languages, frameworks, and databases. Changes or constraints in these environments may impact development timelines and approaches.
- **External Resources:** The availability of external resources, such as design assets, may affect the application's design and aesthetics. Timely access to these resources is crucial for design consistency.
- **User Feedback and Engagement:** The success of the community interaction features depends on user engagement and contributions. The project assumes that users will actively participate in community discussions and content creation.
- **Maintenance and Support:** Ongoing maintenance and support of the application post-launch depend on the availability of resources within the customer's organization. The project assumes adequate support resources will be allocated.

It's important to recognize that these assumptions and dependencies may evolve or change over time. Continuous monitoring and communication with third-party providers, users, and stakeholders will be essential to address any changes that may affect project requirements and outcomes.

3. External Interface Requirements

The "Happy Journey" website version features a set of user interfaces designed to provide an engaging and user-friendly experience to web users. This section outlines the logical characteristics of these interfaces:

- **User Interface Components:**

- **Homepage:** The website's landing page serves as an entry point for users, presenting featured travel destinations, restaurant recommendations, and current weather information. Users can initiate search actions from this page.
- **User Dashboard:** Registered users access personalized dashboards displaying their travel plans, saved attractions, recent bookings, and community activity. This dashboard allows users to manage their activities efficiently.
- **Search Interfaces:** Search functionalities are integrated throughout the website, allowing users to find flights, hotels, restaurants, attractions, and potential travel partners. These interfaces offer search bars, filters, and location-based search options.
- **Restaurant Listings:** When users explore restaurant options, they encounter listings with images, ratings, cuisine types, and user reviews. The website ensures that restaurant details are easily accessible.
- **Attraction Details:** Detailed attraction pages provide users with images, descriptions, visitor reviews, location maps, and nearby attractions. These pages offer comprehensive information for users' decision-making.
- **Booking Flows:** Travel booking interfaces guide users through the process of selecting travel dates, reviewing options, providing traveler information, and making reservations seamlessly.
- **Community Interaction:** Community interaction interfaces encompass discussion boards, comment sections, and platforms for sharing travel experiences, images, and tips with the online community.
- **User Profile:** Each user's profile page displays personal information, travel history, contributions to the community, and settings for preferences and notifications.

- Common User Interface Characteristics:
 - Responsive Design: The website is designed to be responsive, ensuring a consistent and engaging user experience across various devices and screen sizes.
 - Consistent Navigation: The website maintains consistent navigation elements, making it easy for users to access key features from any page.
 - Visual Style: The website adheres to a visually appealing style that follows the established user interface style guide, encompassing a harmonious color scheme, typography, and visual elements.
 - Error Handling: User interfaces include standardized error messages and validation checks to provide clear feedback in case of input errors.
 - Help and Support: A dedicated help center and support feature are accessible from every page, offering assistance, FAQs, and contact options for users.

Detailed design specifications, including wireframes and visual design mock-ups, are documented in a separate User Interface Specification to provide clear guidance to the development and design teams.

3.2 Software Interfaces

The "Happy Journey" website interfaces with various software components, databases, and external services to provide its functionality. This section outlines the connections between this product and other specific software components, including data exchange and communication details.

- Database Integration:

Database System: MySQL, Version 8.0

Purpose: The website stores user profiles, booking information, community content, and other relevant data in the MySQL database.

- Operating System and Hosting:

Operating System: Windows Server, Version [Specify the version, e.g., Windows Server 2019]

Purpose: The website is hosted on a Windows server, ensuring compatibility with the LAMP (Linux, Apache, MySQL, PHP) stack.

Web Technologies and Frameworks:

- Web Server: Apache HTTP Server, Version 2.4

Purpose: Apache serves web pages and handles HTTP requests from users' browsers.

- Backend Framework: Node.js with Express, Version 14.17

I Purpose: Node.js and Express handle server-side operations, including routing and API interactions.

- Frontend Framework: React.js, Version 17

Purpose: React.js powers the website's dynamic and responsive user interfaces.

- API Integrations:

- Travel Booking API: Provider Rapid API

Purpose: The website integrates with Provider RapidAPI's API for real-time flight and hotel booking.

- Real-time Weather API: Google API

Purpose: Real-time weather updates are obtained from the GoogleAPI to offer accurate weather information.

- Data Exchange and Messages:

HTTP/HTTPS: All communication between the website and external services occurs over HTTP/HTTPS for security and data integrity.

Shared Data:

User Authentication: The website shares user authentication and login status with the user database to validate user actions.

User-Generated Content: User-generated content, such as reviews and travel recommendations, is shared and stored within the database.

- Implementation Constraints:

Security Protocols: All data exchanges, especially sensitive user information, must adhere to industry-standard security protocols and encryption to ensure data protection.

API Documentation: Any changes or updates to third-party APIs must be monitored, and the website's integrations must align with the latest API documentation.

The "Happy Journey" website relies on these software interfaces and connections to deliver its features and functionality seamlessly. Integration with external services and databases allows users to access real-time data and relevant information for their travel needs.

3.3 Communications Interfaces

The "Happy Journey" website relies on various communication functions to provide a seamless user experience. This section outlines the communication requirements and protocols associated with the product:

- Email Communication:

Purpose: The website employs email communication for user registration, password reset, booking confirmations, and notifications.

Message Format: Emails are sent in HTML format to ensure readability and proper rendering.

Communication Standards: The website uses the SMTP (Simple Mail Transfer Protocol) for sending email messages.

- Web Browser Communication:

Purpose: Users access the website through web browsers (e.g., Chrome, Firefox, Safari, Edge).

Compatibility: The website is optimized for the latest versions of popular web browsers.

Message Format: Web pages are served in HTML and adhere to web standards (HTML5, CSS3).

Security: Communication between web browsers and the website is secured using HTTPS to encrypt data in transit.

- Network Server Communications:

Purpose: The website communicates with various external servers and APIs to fetch data, such as weather updates and travel booking details.

Protocols: Communication with external servers is over HTTP/HTTPS, adhering to standard RESTful API principles.

Security: Data exchanged with external servers is encrypted using SSL/TLS for secure connections.

- Electronic Forms:

Purpose: Electronic forms are used for user registration, data submission, and content creation.

Form Format: Forms are designed using HTML and JavaScript for dynamic validation and submission.

Data Transfer: User-submitted data is securely transmitted using HTTPS.

- **Data Transfer Rates and Synchronization:**

Data Transfer Rates: The website aims to provide efficient data transfer, with an emphasis on minimizing page load times and real-time data updates.

Synchronization Mechanisms: The website employs asynchronous data fetching to ensure real-time updates without interrupting the user experience.

- **Communication Security and Encryption:**

All communication, including user data transmission and API interactions, is secured through SSL/TLS encryption to protect data in transit.

Sensitive user information, such as login credentials and payment details, is encrypted to ensure data security.

- **Data Privacy:**

The website complies with data privacy regulations (e.g., GDPR) and adheres to strict privacy policies to protect user data.

- **Communication Standards:**

The website follows industry standards for web communication, ensuring interoperability with various systems and services.

3.4 Hardware Interfaces (if applicable):

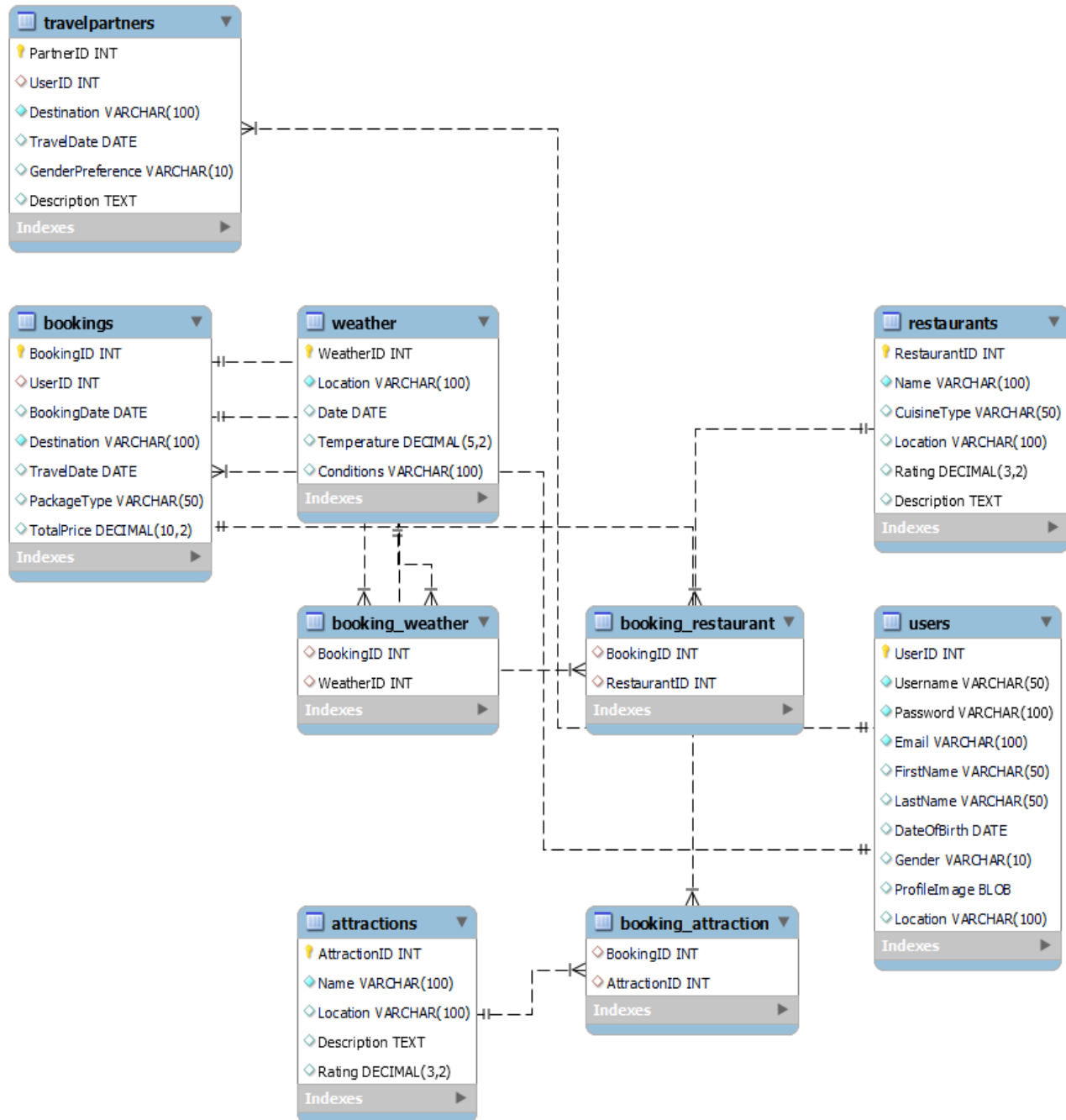
The "Happy Journey" website primarily interfaces with external software components, APIs, and web-based services. It does not require direct hardware interfaces or interactions with

physical devices. The website's functionality is centered around web-based interactions, data exchange, and API integrations.

As the project progresses, if any specific hardware interfaces become necessary for future enhancements or integrations, they will be documented and specified at that time.

4. Analysis Models





5. System Features

This section organizes the functional requirements for the "Happy Journey" website based on major system features. Each feature is described with its priority, stimulus/response sequences, and functional requirements.

System Feature 1: User Registration and Authentication

5.1.1 Description and Priority

Description: Allows users to create accounts, log in, and manage their profiles, including email verification, password reset, and secure authentication.

Priority: High

Benefit: 9

Penalty: 2

Cost: 4

Risk: 3

5.1.2 Stimulus/Response Sequences

User Registration:

Stimulus: User clicks the "Sign Up" button.

Response: The registration form is displayed, prompting the user to provide their details.

User Login:

Stimulus: User enters their login credentials and clicks the "Log In" button.

Response: The system validates the credentials and grants access to the user dashboard.

5.1.3 Functional Requirements

REQ-1: Registration Form

Description: The system must provide a registration form that collects user information, including name, email, and password.

Response to Errors: In case of incomplete or invalid form submission, appropriate error messages must be displayed.

REQ-2: Email Verification

Description: After registration, the system must send a verification email to the user's provided email address.

Response to Errors: In case of email delivery failures, users should be able to request a resend of the verification email.

REQ-3: Password Reset

Description: Users should have the ability to reset their passwords by clicking a "Forgot Password" link.

Response to Errors: When initiating a password reset, the system must guide users through the process with clear instructions.

REQ-4: Secure Authentication

Description: The system must securely store and validate user passwords. User sessions should be maintained securely, and authentication tokens should expire after a defined time.

Response to Errors: The system should handle authentication failures gracefully and inform users about the reasons for access denial.

System Feature 2: Travel Booking

5.2.1 Description and Priority

Description: Enables users to search, compare, and book flights, hotels, and travel packages. Provides real-time pricing and availability information.

Priority: High

Benefit: 8

Penalty: 3

Cost: 5

Risk: 2

5.2.2 Stimulus/Response Sequences

Flight Booking:

Stimulus: User enters travel details and clicks the "Search Flights" button.

Response: The system displays available flight options, prices, and booking details.

Hotel Booking:

Stimulus: User enters destination and dates, then clicks the "Search Hotels" button.

Response: The system lists available hotels with pricing and booking options.

5.2.3 Functional Requirements

REQ-1: Vehicle Search

Description: Users should be able to search for vehicles based on departure/arrival locations, dates, and passenger details.

Response to Errors: The system should handle search errors, such as invalid dates or locations, and provide informative feedback.

REQ-2: Hotel Search

Description: Users can search for hotels based on location, check-in/out dates, and room preferences.

Response to Errors: The system should guide users through the search process and handle errors gracefully.

REQ-3: Price Comparison

Description: The system must compare prices from different airlines and hotels to help users find the best deals.

Response to Errors: In case of price data unavailability, the system should notify users and provide alternatives.

REQ-4: Booking and Reservation

Description: Users should be able to book flights and hotels, including seat selection, payment, and reservation confirmation.

Response to Errors: The system should securely process payments and provide booking confirmation.

System Feature 3: Attractions and Activities

5.3.1 Description and Priority

Description: Provides information about tourist attractions and activities at various destinations. Users can explore and plan their itineraries.

Priority: High

Benefit: 8

Penalty: 2

Cost: 4

Risk: 3

5.3.2 Stimulus/Response Sequences

Attraction Search:

Stimulus: User enters a destination and clicks the "Search Attractions" button.

Response: The system displays a list of attractions, along with details and user reviews.

Itinerary Planning:

Stimulus: User selects attractions and adds them to their itinerary.

Response: The system helps users plan their trips, including scheduling attractions and generating itineraries.

5.3.3 Functional Requirements

REQ-1: Attraction Information

Description: The system must provide comprehensive information about tourist attractions, including descriptions, opening hours, and visitor reviews.

Response to Errors: In case of missing data, the system should inform users about the unavailability of specific details.

REQ-2: User Reviews

Description: Users should be able to write and read reviews for attractions.

Response to Errors: The system should moderate reviews to filter inappropriate content and handle reporting of problematic reviews.

REQ-3: Itinerary Creation

Description: Users can create and manage their trip itineraries by adding attractions, specifying visit dates, and receiving automated suggestions.

Response to Errors: The system should help users recover from errors when planning their itineraries, like adding unavailable attractions.

System Feature 4: Real-Time Weather Updates

5.4.1 Description and Priority

Description: Provides real-time weather information for the user's selected destination. Offers current conditions, forecasts, and weather alerts.

Priority: High

Benefit: 7

Penalty: 2

Cost: 4

Risk: 3

5.4.2 Stimulus/Response Sequences

Weather Search:

Stimulus: User enters a location and clicks the "Get Weather" button.

Response: The system retrieves and displays real-time weather data for the specified location.

Weather Alerts:

Stimulus: The system detects severe weather conditions.

Response: Users receive alerts and warnings about weather emergencies.

5.4.3 Functional Requirements

REQ-1: Weather Data Retrieval

Description: The system should fetch real-time weather data from reliable sources, including temperature, humidity, wind speed, and forecasts.

Response to Errors: In case of data unavailability or retrieval errors, the system should notify users and provide alternative sources.

REQ-2: User Location

Description: The system must allow users to specify their current location or a destination for weather updates.

Response to Errors: In case of location input errors, the system should guide users to correct entries.

REQ-3: Weather Alerts

Description: The system should monitor weather conditions and send alerts to users in case of severe weather events.

Response to Errors: The system should provide clear, real-time alerts and guidance on safety measures in emergencies.

System Feature 5: Community Interaction

5.5.1 Description and Priority

Description: Enables users to interact with a community of fellow travelers. They can share experiences, ask questions, and provide recommendations.

Priority: Medium

Benefit: 6

Penalty: 3

Cost: 4

Risk: 4

5.5.2 Stimulus/Response Sequences

Posting a Travel Experience:

Stimulus: User writes a post about their recent travel experience.

Response: The system displays the post on the community board for other users to read and comment on.

Asking Questions:

Stimulus: User asks a question regarding their upcoming trip.

Response: The system posts the question, and other users can respond with answers and advice.

5.5.3 Functional Requirements

REQ-1: Community Board

Description: The system must provide a community board where users can post their travel experiences, questions, and recommendations.

Response to Errors: In case of posting errors or inappropriate content, the system should provide moderation and reporting features.

REQ-2: User Interaction

Description: Users should be able to like, comment on, and share posts. They can also follow other users.

Response to Errors: The system should ensure user comments and interactions adhere to community guidelines.

REQ-3: Notifications

Description: The system should send notifications to users when someone interacts with their posts, responds to their questions, or follows them.

Response to Errors: Notifications should be accurate and timely, and the system should handle cases where notifications fail.

System Feature 6: Travel Partner Search

5.6.1 Description and Priority

Description: Enables users to search for travel partners based on shared interests, travel plans, and preferences.

Priority: Medium

Benefit: 6

Penalty: 3

Cost: 4

Risk: 4

5.6.2 Stimulus/Response Sequences

Search for Travel Partners:

Stimulus: User enters their travel plans and interests and clicks the "Find Travel Partners" button.

Response: The system displays a list of potential travel partners who match the user's criteria.

Request to Join Travel Plans:

Stimulus: A user sends a request to join someone else's travel plans.

Response: The system notifies the user if their request is accepted or declined.

5.6.3 Functional Requirements

REQ-1: Profile Creation

Description: Users should be able to create detailed profiles, including travel preferences and interests.

Response to Errors: The system should guide users through profile creation and handle errors during profile setup.

REQ-2: Search Algorithm

Description: The system must implement an effective search algorithm to match users with similar travel plans and interests.

Response to Errors: In case of search algorithm issues, the system should provide alternative search options and accurate matching.

REQ-3: Communication

Description: Users should be able to communicate with potential travel partners through a messaging system.

Response to Errors: The system should ensure secure and reliable communication between users.

System Feature 7: Travel Booking

5.7.1 Description and Priority

Description: Allows users to book travel-related services such as flights, hotels, and travel packages directly through the application.

Priority: High

Benefit: 8

Penalty: 2

Cost: 5

Risk: 3

5.7.2 Stimulus/Response Sequences

Cab Booking:

Stimulus: User selects a destination, travel dates, and cab preferences.

Response: The system displays available flights and allows the user to make a booking.

Hotel Reservation:

Stimulus: User enters check-in/check-out dates and hotel preferences.

Response: The system shows available hotels and facilitates reservations.

5.7.3 Functional Requirements

REQ-1: Booking Options

Description: The system must integrate with external travel booking services to offer users a variety of options for flights, hotels, and travel packages.

Response to Errors: In case of booking failures or unavailability, the system should provide alternatives or prompt users to retry.

REQ-2: Secure Payment

Description: Users should be able to make secure payments for bookings within the application.

Response to Errors: The system should ensure the security of user payment information and handle payment processing errors.

REQ-3: Booking Confirmation

Description: Users should receive booking confirmations and itineraries after successful bookings.

Response to Errors: The system should provide clear booking statuses and guidance for users in case of booking issues.

6. Other Nonfunctional Requirements

6.1 Performance Requirements

Performance requirements define how well the system should perform under various conditions. They are crucial to ensure that the system meets user expectations and functions optimally. Specific performance requirements should be defined for individual functional requirements or features, as necessary.

- Response Time:

The system should respond to user actions (e.g., search queries) within 2 seconds under normal load conditions.

- Scalability:

The system should be able to handle a minimum of 10,000 concurrent users without degradation in response time.

- Data Retrieval:

Data retrieval from external services (e.g., real-time weather updates) should have a latency of no more than 5 seconds.

- Peak Load Handling:

During peak usage periods (e.g., holiday seasons), the system should maintain response times below 3 seconds and be capable of handling 50,000 concurrent users.

- Error Handling:

The system should provide informative error messages within 5 seconds of encountering an error to prevent user frustration.

Performance requirements are driven by the need to provide a seamless and efficient user experience. Meeting these requirements ensures that users can access the system, retrieve data, and perform actions without undue delays.

6.2 Safety Requirements

Safety requirements are concerned with preventing potential harm or damage that could result from the use of the product. They define actions that must be taken to ensure user safety, as well as actions that must be prevented. Compliance with external policies and regulations that affect safety should also be addressed.

- Data Security:

All user data, including personal and payment information, must be stored securely and encrypted to prevent unauthorized access.

- User Privacy:

The system should comply with relevant data protection regulations (e.g., GDPR) to safeguard user privacy.

- **Emergency Information:**

The application must provide access to emergency information such as local emergency contact numbers and healthcare facilities when users are traveling.

- **No Harmful Content:**

The system should prevent the display of harmful or inappropriate content in user-generated posts or recommendations.

- **Safety Certifications:**

The application should meet industry safety standards and certifications, as applicable.

- **Data Backup:**

Regular data backups should be performed to prevent data loss due to system failures or errors.

These safety requirements aim to protect users and ensure that the application is safe to use. Compliance with relevant policies and industry standards is essential to prevent potential harm or damage.

6.3 Security Requirements

Security requirements are vital to address security and privacy issues surrounding the product. They define measures to protect user data and ensure data privacy. Compliance with external policies and regulations affecting security should be considered.

- **User Authentication:**

Users must be required to authenticate their identities through secure login methods, such as usernames and passwords.

- Data Encryption:

All user data, including personal information and payment details, must be encrypted during storage and transmission.

- Access Control:

The system should implement role-based access control to ensure that users can access only the data and features relevant to their roles.

- Data Privacy Compliance:

The system must comply with data protection regulations, such as GDPR, and ensure user data privacy.

- Secure Communication:

Secure communication protocols (e.g., HTTPS) should be used to protect data during transmission.

- Security Audits:

Regular security audits and vulnerability assessments should be conducted to identify and address security risks.

- Incident Response:

A documented incident response plan should be in place to address security breaches and data breaches promptly.

- Security Certifications:

The application should meet relevant security certifications and standards, as applicable. These security requirements are essential to safeguard user data and ensure that the application is secure for use. Compliance with data protection regulations and industry security standards is crucial.

6.4 Software Quality Attributes

Software quality attributes define additional quality characteristics that are important to customers or developers. These attributes should be specific, quantitative, and verifiable whenever possible.

- Usability:

The application should have an average user satisfaction rating of at least 4.5 out of 5 based on user feedback.

- Reliability:

The system should have a 99.99% uptime guarantee, with no more than 1 hour of planned downtime annually.

- Maintainability:

Any software updates or maintenance tasks should not result in more than 10 minutes of system unavailability.

- Performance:

The system should load pages and data within 2 seconds for 95% of user requests.

- Robustness:

The application should handle unexpected inputs or errors gracefully and provide informative error messages.

- Portability:

The application should be accessible and function well on major web browsers (e.g., Chrome, Firefox, Safari) and devices (desktop, tablet, mobile).

- Interoperability:

The application should integrate with popular third-party APIs (e.g., weather data providers) without issues.

These software quality attributes address various aspects of the product's quality and performance. Quantitative measures help ensure that these attributes are met.

6.5 Business Rules

This section outlines the operating principles and business rules governing the use of the "Happy Journey" application. These rules specify which individuals or roles can perform particular functions under specific circumstances.

- User Roles:

The application defines three primary user roles: "Traveler," "Restaurant Owner," and "Administrator." Each role has specific permissions and capabilities.

- User Authentication:

Users must authenticate their identities using unique usernames and passwords. Two-factor authentication is encouraged for added security.

- User Reviews:

Travelers can submit reviews and ratings for restaurants and attractions. Reviews must adhere to community guidelines, including no offensive or inappropriate content.

- Restaurant Listings:

Only authorized restaurant owners can create and manage restaurant listings. Listings should be accurate and up-to-date.

- Administrator Functions:

Administrators have the authority to monitor and moderate user-generated content, enforce community guidelines, and manage user accounts.

- Data Privacy:

The application must adhere to data protection regulations, including GDPR, to ensure user data privacy.

- Payment Processing:

Payment processing for bookings is handled securely in compliance with PCI DSS and other industry standards.

- Search Algorithms:

The search algorithm should prioritize search results based on user preferences, location, and relevance, as defined by industry-specific guidelines.

- Geo-location Data:

The product uses geo-location data for location-based features but only with user consent.

- Compliance:

The application complies with all relevant industry standards, legal requirements, and community guidelines.

These business rules ensure the responsible use of the application, define user roles and their capabilities, and provide guidelines for content creation and data privacy. Compliance with these rules is essential to maintain a safe and ethical environment for users.

6.5 Domain Requirements

Domain requirements are industry-specific regulations and standards that your website must adhere to. These requirements help ensure that the website operates within the guidelines of your specific domain.

- Industry Regulations:

The website must comply with all relevant industry regulations and standards. For example, if the website is related to travel and tourism, it should adhere to industry-specific regulations governing booking platforms and travel agencies.

- Content Guidelines:

Content on the website, including user-generated content, must adhere to industry-specific guidelines and standards. For instance, if the website deals with health and wellness, content should comply with regulations related to medical information and wellness advice.

- Data Security:

If your domain involves sensitive data, the website must implement data security practices in line with industry standards.

- Accessibility:

The website should be accessible to users with disabilities in accordance with industry guidelines, such as WCAG (Web Content Accessibility Guidelines).

- Search Algorithms:

If your domain has specific search requirements (e.g., e-commerce, job search), the search algorithm should meet industry-specific standards for relevance and accuracy.

- Industry-Specific Integration:

If your website integrates with industry-specific third-party services or APIs (e.g., payment gateways, industry-specific data providers), it should do so in compliance with industry standards.

7. Other Requirements:

This section covers additional requirements that may not fit within the previous sections of the SRS. These requirements encompass a range of factors that impact the project.

- Database Requirements:

The website shall use a relational database management system (DBMS) to store user data, content, and transaction records. Specific database design and schema details will be provided in a separate document.

- Internationalization:

The website should be designed with internationalization in mind, allowing for the potential expansion into different languages and regions. This includes support for multiple languages and localization of content.

- Legal Requirements:

The website shall adhere to all relevant national and international laws and regulations, including but not limited to data protection, copyright, and accessibility laws. Legal compliance is a top priority.

- Reuse Objectives:

The project shall aim to create reusable components, such as code libraries and design assets, for potential use in future projects. These components should be documented for easy integration.

- Backup and Recovery:

Regular data backups shall be performed to ensure data integrity and availability. A disaster recovery plan should be in place to minimize downtime in case of unexpected events.

- Performance Testing:

The project should undergo performance testing to ensure that the website can handle expected loads. Specific performance criteria and testing procedures will be defined in a separate document.

- Browser Compatibility:

The website must be compatible with a range of modern web browsers, including but not limited to Chrome, Firefox, Safari, and Edge.

- Mobile Responsiveness:

While not a primary focus, the website should be responsive and functional on mobile devices to accommodate users who access it from smartphones and tablets.

- Documentation:

Comprehensive documentation for the website's architecture, code, and usage should be created for the benefit of developers, administrators, and future maintainers. These additional requirements cover various aspects that impact the project's success, including technical considerations, legal compliance, and future development possibilities.

Appendix A: Glossary

This glossary provides definitions for terms, acronyms, and abbreviations used throughout this Software Requirements Specification (SRS). It is intended to assist in interpreting the document.

SRS: Software Requirements Specification - A document that outlines the functional and non-functional requirements of a software project.

API: Application Programming Interface - A set of rules and protocols that allow different software applications to communicate with each other.

UI: User Interface - The graphical or textual elements through which users interact with a software application.

DBMS: Database Management System - Software that manages and organizes data in a database.

GDPR: General Data Protection Regulation - European Union regulations regarding data protection and privacy.

PCI DSS: Payment Card Industry Data Security Standard - A set of security standards for organizations that handle credit card transactions.

WCAG: Web Content Accessibility Guidelines - A set of guidelines for making web content more accessible to people with disabilities.

SEO: Search Engine Optimization - The practice of optimizing a website to improve its visibility in search engine results.

CMS: Content Management System - A software application for creating, managing, and modifying digital content.

URL: Uniform Resource Locator - A reference used to access resources on the internet, often in the form of a web address.

HTML: Hypertext Markup Language - The standard language for creating web pages.

CSS: Cascading Style Sheets - A style sheet language used for describing the presentation of a document written in HTML.

JavaScript: A programming language commonly used to create interactive effects within web browsers.

UX: User Experience - The overall experience of a person using a product, system, or service.

HTTPS: HyperText Transfer Protocol Secure - The secure version of HTTP, used to encrypt data between the user and the webserver.

SQL: Structured Query Language - A domain-specific language for managing and querying relational databases.

UI/UX Design: User Interface/User Experience Design - The process of designing the look and feel of a website or application to enhance the user experience.

Appendix B: Field Layouts

Field Name	Field Type	Description	Required	Validation Rules
First Name	Text	User's first name	Yes	Alphabetic characters only
Last Name	Text	User's last name	Yes	Alphabetic characters only
Email	Email Address	User's email address	Yes	Valid email format
Password	Password	User's password	Yes	Strong password requirement
Date of Birth	Date	User's date of birth	No	Vaild date format
Profile Picture	Image	User's profile picture	No	Image file format validation
Booking Date	Date	Date of travel booking	Yes	Vaild date format
Booking Amount	Currency	Amount paid for the booking	Yes	Numeric, Positive value only
Phone number	Phone Number	User's contact number	No	vaild phone number format
Address	Text	User's address	No	Text input
Travel Destination	Dropdown	User's selected travel destination	Yes	Predefined options
Payment Method	Radio Button	User's chosen payment method	Yes	Select one option
Trip Duration	Number	Number of days for the trip	Yes	Numeric, positive value only
Booking Reference	Text	Unique reference for the booking	Yes	Alphanumeric characters only

Trip Type	Radio Button	One-way or Round-trip	Yes	Select one option
Special Request	Text Area	User's special requests or notes	Yes	Text input

REPORT REQUIREMENT:

Report Name	Description	Data Source	Filter Options	Display Format
User Profile	Display user details	User database	User ID, date of Birth, etc..	HTML, PDF, CSV
Booking History	List of past travel booking	Booking database	Date range, Location, etc..	HTML, Excel
Revenue Summary	Monthly revenue report	Booking database	Date range, service type, etc.	HTML, PDF
User Activity	User interaction summary	User logs	Date range, activity type, etc..	HTML, Excel
User Feedback	User-provided feedback and ratings	Feedback database	Date range, rating service, etc..	HTML, Excel
Popular Destinations	Most Booked travel destinations	Booking database	Date range, Destination type, etc..	HTML, PDF, CSV
Service Utilization	Summary of service utilization	Booking database	Date range, service type, ect	HTML, PDF

Sample sheet with information required to register the customer

Field	Length(bytes)	Data type	Description	Is Mandatory
Account Number	16	Numeric	Customer's account number	Y

ISFC code	11	Alphanumeric	Indian Financial System Code	Y
Card Amount	20	Numeric	Initial Card Amount	Y
Mandate Start Date	8	Date	Date of Mandate Registration	N
Mandate End Date	8	Date	Date of Mandate Expiry	N
Status	25	Alphanumeric	Status of Registration	Y
Customer Name	60	String	Customer's full Namex`	Y
Reject Reason Code	4	String	Reject Reason code in case mandate is rejected	N

Sample Report Requirements: Include the fields to be included in the report

Registration Report	Transaction Report
Bank Account Number	Transaction Reference Number
ISFC Code	Bank Account Number
Bank Name	IFSC Code
Account Status	Bank Name
Account Type	Customer Name
Customer Name	Card Number
Card Number	Debit Transaction Amount
SI Start Date	Transaction Date

Status	Status
Remarks	Debit Attempt Number
	Remarks

Appendix C: Requirement Traceability matrix

Sl. No.	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID
1	REQ-001	User registration	Arch-001	Design-001	UserRegModule.js	TC-001
2	REQ-002	Search for restaurants	Arch-002	Design-002	RestaurantSearch.js	TC-002
3	REQ-003	Real-time weather updates	Arch-003	Design-003	WeatherModule.js	TC-003
4	REQ-004	Book travel	Arch-004	Design-004	TravelBooking.js	TC-004
5	REQ-005	Share travel experiences	Arch-005	Design-005	SocialModule.js	TC-005
6	REQ-006	Attractions information	Arch-006	Design-006	AttractionsModule.js	TC-006
7	REQ-007	User profile management	Arch-007	Design-007	UserProfileModule.js	TC-007
8	REQ-008	Payment processing	Arch-008	Design-008	PaymentModule.js	TC-008
9	REQ-009	Customer feedback	Arch-009	Design-009	FeedbackModule.js	TC-009
10	REQ-010	Travel partner search	Arch-010	Design-010	PartnerSearch.js	TC-010